DAY 06 NOTES

# Single-Qubit Quantum Gates

Comprehensive Study Guide — Project-Q 30-Day Quantum Challenge

## 1. Foundational Context: Why Quantum Gates Matter

A quantum computer cannot execute algorithms or evaluate mathematical functions without operations to actively modify its internal data structures. Just as classical compute processing hardware uses legacy silicon logic gates to evaluate deterministic bits, quantum processors utilize **Quantum Gates** to apply matrix transformations directly onto the complex state vectors of active qubits.

### Architectural Comparison Matrix

| Operational Metric | Classical Logic Gates | Quantum Gates |
|---|---|---|
| **Target Primitive** | Operates exclusively on discrete classical bits (0 or 1). | Operates on continuous complex state vectors ($\alpha\|0\rangle + \beta\|1\rangle$). |
| **Input Flexibility** | Bound strictly to rigid binary input constants. | Processes any valid coherent superposition state vector. |
| **Physical Logic** | Performs standard Boolean logical operations. | Executes spatial complex state vector transformations. |
| **Information Loss** | **Irreversible:** Structural data is lost upon gate execution (e.g., AND/OR). | **Strictly Reversible:** Every operator has an inverse, fully preserving vector properties. |

## 2. Geometric Interpretation on the Bloch Sphere

Instead of conceptualizing quantum gates as basic digital switches (such as changing a classical 0 directly to a 1), they are formally evaluated as **geometric rotations** of the state vector around fixed, orthogonal spatial coordinates on a unit three-dimensional sphere, known as the **Bloch Sphere**. Every single-qubit manipulation maps cleanly to an angular sweep along a specified axis.

## 3. Mathematical Deep-Dive: Core Single-Qubit Operators

### A. The Pauli-X Gate (The Quantum NOT / Bit-Flip)

The Pauli-X gate serves as the quantum analogue to a classical inverter circuit. When applied to baseline states, it swaps the probability amplitudes cleanly.

$$\textit{Matrix Formulation:} \quad X = [\ 0 \quad 1\ ;\ 1 \quad 0\ ]$$

$$\textit{State Mapping:} \quad X|0\rangle = |1\rangle \quad | \quad X|1\rangle = |0\rangle$$

- **Vector Geometry:** Rotates the state vector by exactly *180^\circ* ($\pi$ radians) around the horizontal **X-axis**. This operation sweeps the North Pole ($|0\rangle$) to the South Pole ($|1\rangle$) and vice-versa.
- **Arbitrary Action:** $X(\alpha|0\rangle + \beta|1\rangle) = \beta|0\rangle + \alpha|1\rangle$ (Swaps operational scalar coefficients).

### B. The Pauli-Y Gate (Bit-Flip + Phase-Flip)

The Pauli-Y gate performs a simultaneous transposition of the computational basis configurations alongside an internal phase migration.

$$\textit{Matrix Formulation:} \quad Y = [\ 0 \quad -i\ ;\ i \quad 0\ ]$$

$$\textit{State Mapping:} \quad Y|0\rangle = i|1\rangle \quad | \quad Y|1\rangle = -i|0\rangle$$

**The inclusion of the imaginary unit $i$ explicitly indicates the injection of an absolute phase translation.*

- **Vector Geometry:** Initiates a sharp *180^\circ* rotation directly around the spatial **Y-axis** of the sphere.

### C. The Pauli-Z Gate (The Phase-Flip Operator)

Unlike the X and Y configurations, the Pauli-Z gate leaves the baseline ground state $|0\rangle$ completely unaltered. It targets the excited state $|1\rangle$ with a relative sign inversion.

$$\textit{Matrix Formulation:} \quad Z = [\ 1 \quad 0\ ;\ 0 \quad -1\ ]$$

$$\textit{State Mapping:} \quad Z|0\rangle = |0\rangle \quad | \quad Z|1\rangle = -|1\rangle$$

- **Vector Geometry:** Induces a complete *180^\circ* rotation around the vertical, polar **Z-axis**.
- **Algorithmic Significance:** While immediate measurement percentages remain identical ($|-1|^2 = 1$), modifying this relative phase alters how the wave-vector combines with downstream gates, enabling constructive or destructive computational interference.

### D. The Hadamard (H) Gate (The Superposition Engine)

The Hadamard gate is the cornerstone for initializing quantum parallelism. Its primary function is transforming baseline, definite states into an balanced, equal superposition state vector.

$$\textit{Matrix Formulation:} \quad H = (1/\sqrt{2}) * [\ 1 \quad 1\ ;\ 1 \quad -1\ ]$$

$$H|0\rangle = (|0\rangle + |1\rangle)\ /\ \sqrt{2} \quad \equiv \quad |+\rangle$$

$$H|1\rangle = (|0\rangle - |1\rangle)\ /\ \sqrt{2} \quad \equiv \quad |-\rangle$$

- **Vector Geometry:** Rotates the state vector by *180^\circ* around a diagonal axis running exactly midway between the core X and Z coordinate nodes, dropping the polar vector flat onto the equatorial plane.
- **Algorithmic Inception:** This gate acts as the standard premier layout rule for deep parallelism algorithms including Grover's Search, Shor's Factoring, and the Quantum Fourier Transform (QFT).

## 4. Master Control Reference Matrix

| Gate | Primary State Action | Bloch Sphere Rotation | Alters Phase? | Matrix Form |
|---|---|---|---|---|
| **X** | Flips computational basis states ($0 \leftrightarrow 1$) | 180° around the X-axis | No | [0, 1; 1, 0] |
| **Y** | Flips basis states + Rotates relative phase | 180° around the Y-axis | Yes | [0, -i; i, 0] |
| **Z** | Inverts sign tracking of state $\|1\rangle$ selectively | 180° around the Z-axis | Yes | [1, 0; 0, -1] |
| **H** | Maps fixed binary states into equal superposition | 180° around diagonal X-Z axis | Yes | 1/√2 * [1, 1; 1, -1] |

## 5. Deconstructing Core Concept Misconceptions

⚠️ **Misconception 1: "The X gate acts exactly like a classical logic NOT inverter under all possible criteria."**
**Correction:** This rule is true only if the target qubit is resting purely inside a baseline state ($|0\rangle$ or $|1\rangle$). If it encounters a complex superposition vector, it transposes the scalar coefficients amplitude positioning instead of flipping a binary flag.

⚠️ **Misconception 2: "The Pauli-Z gate does nothing useful because the calculated absolute squared measurement odds do not change."**
**Correction:** While immediate measurement odds are unchanged, the phase signature alteration drastically impacts how that state interacts with subsequent gates further down the execution wire, acting as the bedrock for algorithmic wave interference.

⚠️ **Misconception 3: "A Hadamard gate causes the qubit to return both 0 and 1 as active simultaneous outcomes upon measurement reading."**
**Correction:** The H gate prepares a smooth mathematical *superposition wave vector*. The moment a physical measurement tool interacts with that vector, it instantly triggers state collapse, forcing a singular, definitive classical result (0 **OR** 1).